

# Spectroscopy of unbound $^{17}\text{C}$ states

via  $^{16}\text{C}(\text{d},\text{p})^{17}\text{C}$  at FRIB (a1975)

*Angular distributions · absolute normalization · L assignments*

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spectroscopic reference: Movilla & Ayyad (2026-02)

- $^{16}\text{C}$  beam in the AT-TPC,  $\text{D}_2$  active target (FRIB a1975).
- 188.33 MeV total = 11.77 MeV/u;  $Q_{g.s.} = -1.498$  MeV.
- Transfer populates the bound region + seven unbound resonances.
- Goal: per-state  $d\sigma/d\Omega$  on an absolute scale
  - $\rightarrow$  transferred- $L$  from FRESKO ADWA shape comparison.
- Data reduced with ATTPCROOT (InterpSolver); cont. of C16dp\_fits.

## Unbound resonances fitted

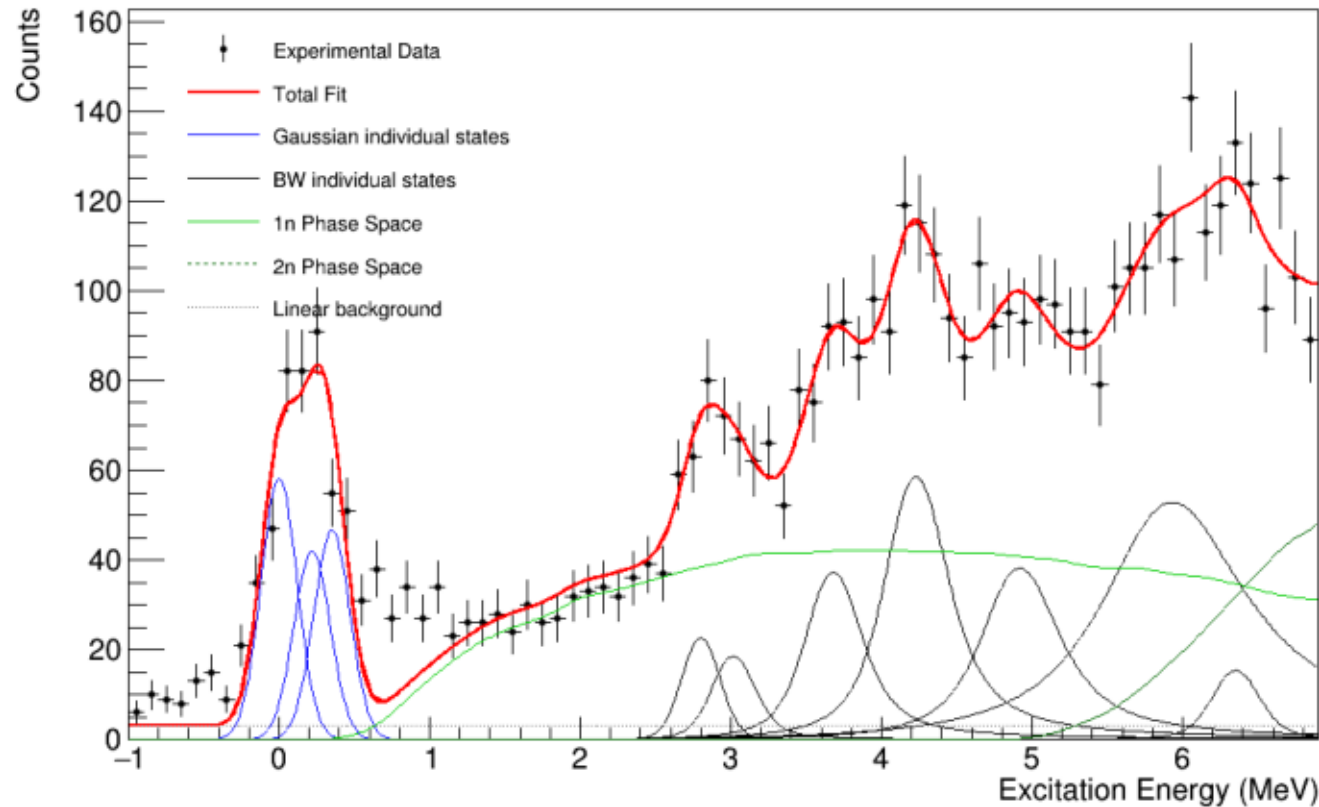
| Ex (MeV) | Jpi  | Gamma (keV) |
|----------|------|-------------|
| -----    |      |             |
| 2.763    | 1/2- | 50          |
| 2.980    | 3/2+ | 200         |
| 3.661    | new  | 398         |
| 4.231    | 3/2+ | 500         |
| 4.841    | 1/2+ | 300         |
| 5.91     | 3/2+ | 1500        |
| 6.30     | 5/2+ | 396         |

- 1. Spectral fit (C16\_pd\_AngBins.C):  $12 \times 2.5^\circ$  bins,  $10\text{--}40^\circ$ .
  - 3 Gaussians (bound) + 7 penetrability-modified Breit-Wigners
  - +  $1n/2n$  phase space + linear bg; analytic Voigt lineshape.
  - Inclusive fit pins positions/widths; per-bin frees amplitudes.
- 2. Efficiency: 8 Ex-tables, bilinear ( $E_x, \theta$ ); floor+gradient cuts.
- 3. Absolute normalization fixed by the  $^{16}\text{C}(d,d)$  elastic channel.
- 4. L scan: each AD vs FRESCO  $L=0\text{--}3$  ADWA shapes; best L by  $\chi^2/\nu$ .
- 5. Systematics: 82-variant campaign  $\rightarrow$  per-state envelope.

# Inclusive $E_x$ spectrum: the global fit

spectral fit

Inclusive 10-40 deg



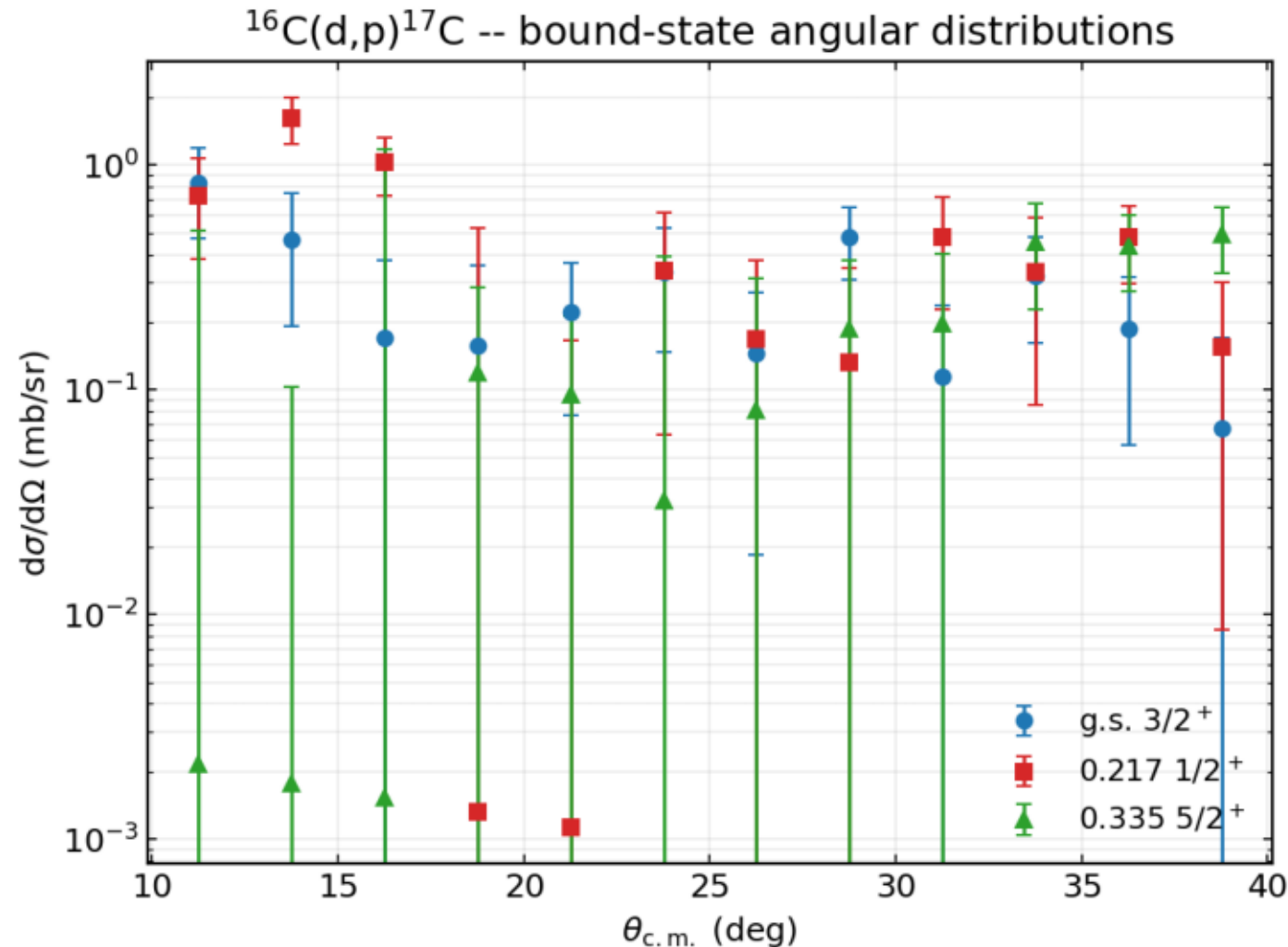
- A localized  $+2\text{--}4\sigma$  excess near 0.4–1.1 MeV is not absorbed by the nominal model.
- Adding a contaminant Gaussian (centroid pegs  $\sim 0.45$  MeV) restores an acceptable fit:

| cuts   | nominal   | +contam |
|--------|-----------|---------|
| -----  |           |         |
| loose  | 1.78      | 0.84    |
| strict | 1.88      | 0.97    |
|        | (chi2/nu) |         |

- Survives strict cuts (0.450  $\rightarrow$  0.451 MeV)  $\Rightarrow$  not an AT-TPC edge artifact.
- Carried as a model-dependence systematic.

# Bound states: angular distributions

comparison



- Three bound  $^{17}\text{C}$  states, as a cross-check of the machinery:

- g.s.  $3/2^+$

- 0.217 MeV  $1/2^+$

- 0.335 MeV  $5/2^+$

- Forward-peaked g.s. and 0.217 shapes consistent with the expected  $\ell$  content.

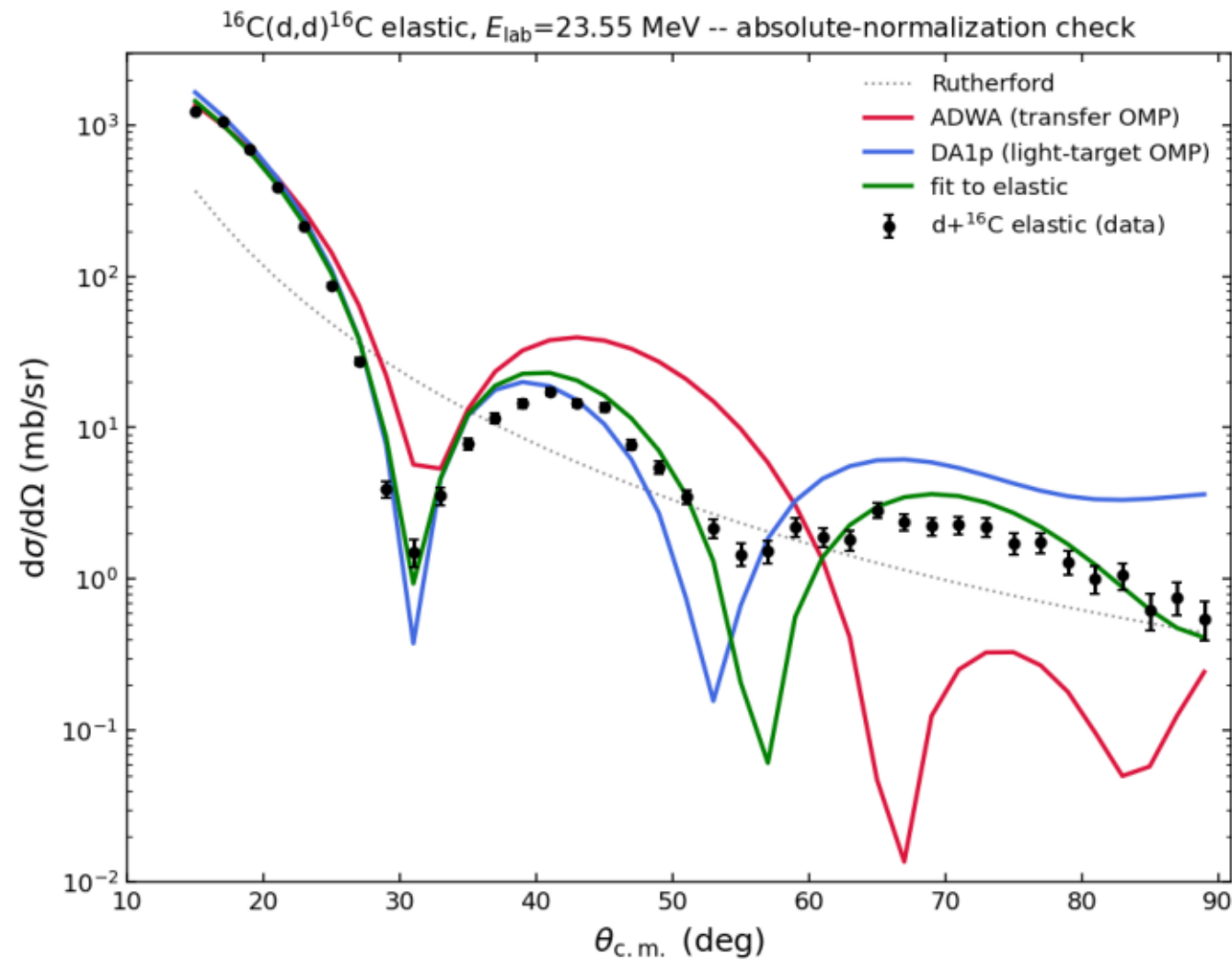
- The  $5/2^+$  is only weakly populated — its points sit near the detection floor (large bars), shown for completeness.

- Validate efficiency + absolute scale before the unbound region.

Counts → cross section (fixed luminosity constant):

$$\frac{d\sigma}{d\Omega} = \frac{Y}{N_{beam} N_{tgt} \Delta\Omega} \cdot 10 / \varepsilon$$

- $N_{beam} = 1.6146 \times 10^5$ ,  $N_{tgt} = 0.019632$ ,  $\Delta\Omega = 2\pi(\cos \theta_{lo} - \cos \theta_{hi})$ .
- Cross-check: the  $^{16}\text{C}(d,d)$  elastic channel shares the entrance partition (same beam,  $\text{D}_2$  target, luminosity).
  - An absolute OM elastic cross section that reproduces the data validates the constant — no free normalization.
- Self-contained partial-wave OM solver (om\_elastic.py): Numerov + Coulomb matching; vs Rutherford to  $10^{-3}$ .
  - $E_{lab}(d) = 23.55 \text{ MeV}$ ,  $E_{c.m.} = 20.92 \text{ MeV}$ .



## Forward-lobe (data/OM):

| OMP             | chi2/nu | d/OM  |
|-----------------|---------|-------|
| -----           |         |       |
| ADWA (transfer) | 398     | 0.864 |
| DA1p (global)   | 105     | 0.886 |
| free 5-par fit  | 32      | 0.978 |

- Absolute scale consistent with unity — no rescale.
- BUT the transfer ADWA potential itself sits ~14% low.
- Honest figure = cross-potential spread  $\approx 0.86\text{--}0.98$  (~12%), a GLOBAL scale uncertainty (not in per-point errors).

$$\delta\left(\frac{d\sigma}{d\Omega}\right) = \frac{d\sigma}{d\Omega} \sqrt{(\delta Y/Y)^2 + (\delta \varepsilon/\varepsilon)^2}$$

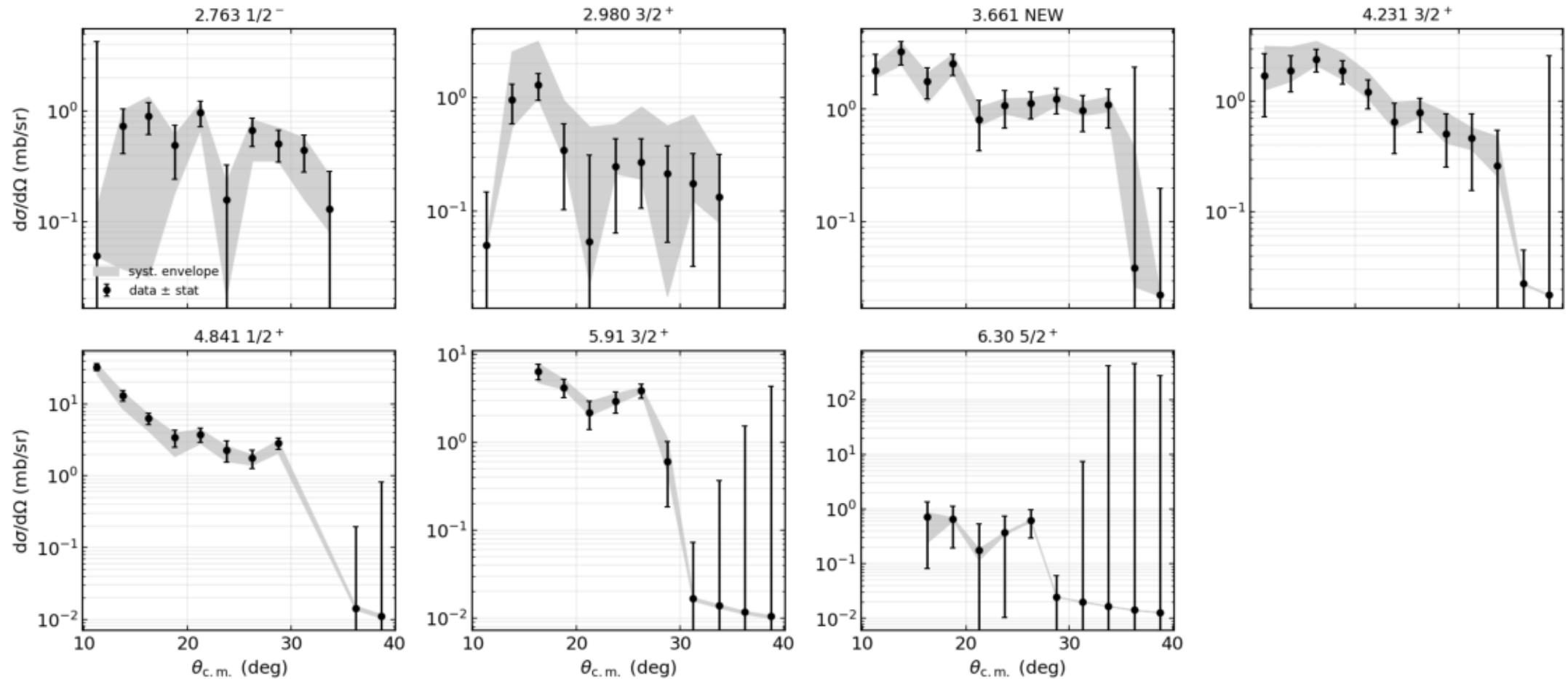
- $\delta Y$  = Minuit amplitude error ( $\chi^2$  fit to  $\sqrt{N}$ -weighted histograms) — already carries Poisson stats.
- FIXED (2026-05-30): a previous  $\sqrt{1/Y + (\delta Y/Y)^2}$  double-counted Poisson ( $\sqrt{2}$  inflation).
  - e.g. g.s. 10-12.5°: 0.433 → 0.356 mb/sr.
- Three uncertainty layers, kept separate:
  - statistical (above) + systematic (82-variant envelope) + global scale (~12% OMP spread).



# Unbound resonances: $d\sigma/d\Omega$ + systematic band

*angular distributions*

$^{16}\text{C}(d,p)^{17}\text{C}$  unbound resonances --  $d\sigma/d\Omega$  with 82-variant systematic band



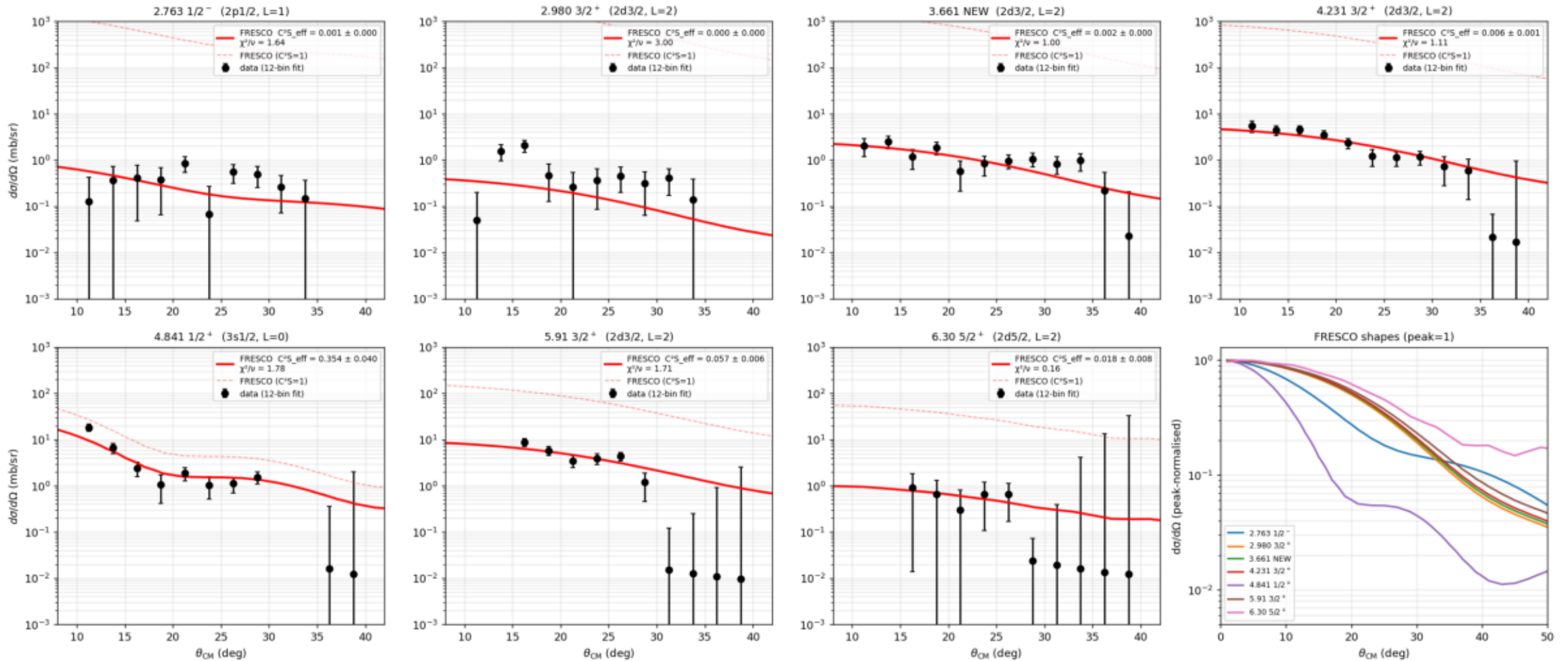
Black: data  $\pm$  corrected statistical error. Gray: min-max across the 82 spectral-fit variants. Fits use  $\theta_{c.m.} \leq 30^\circ$ .

- FRESCO 7-state DWBA with the Johnson–Soper ADWA effective deuteron OMP (sum of Koning–Delaroche  $p + {}^{16}\text{C}$  and  $n + {}^{16}\text{C}$  at  $E_d/2$ ).
- Four templates  $L=0,1,2,3$  (neutron in  $3s_{1/2}, 2p_{1/2}, 2d_{3/2}/2d_{5/2}, 1f_{5/2}$ ); weakly-bound approx  $b_e=0.5$  MeV.
- Each AD fit to each L shape (single normalization); best L by  $\chi^2/\nu$  on  $\theta_{c.m.} \leq 30^\circ$ .
- SHAPE-ONLY: the WBA reproduces the L-dependent shape but not the magnitude  $\Rightarrow C^2S$  are effective, not publication SFs
  - (needs CDCC / pole-residue treatment).

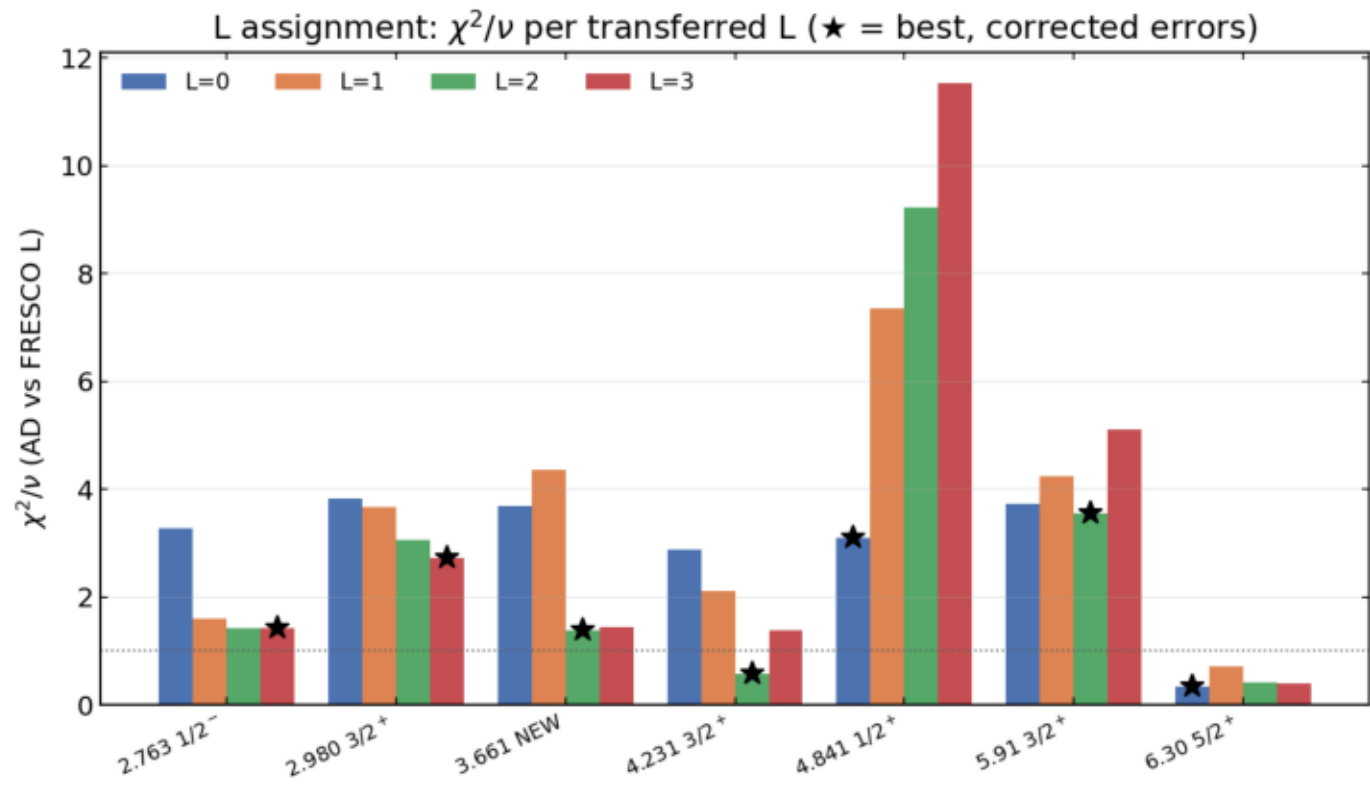
# L assignment: FRESCO shape overlays

*L values*

16C(d,p)17C @ 11.77 MeV/u — FRESCO ADWA (weakly-bound approx) vs extracted AD



Curves = FRESCO ADWA at each state's literature single-particle config (normalization-fit, full 10–40°). The data-driven all- $L$  scan on  $\theta_{c.m.} \leq 30^\circ$  is the next slide.

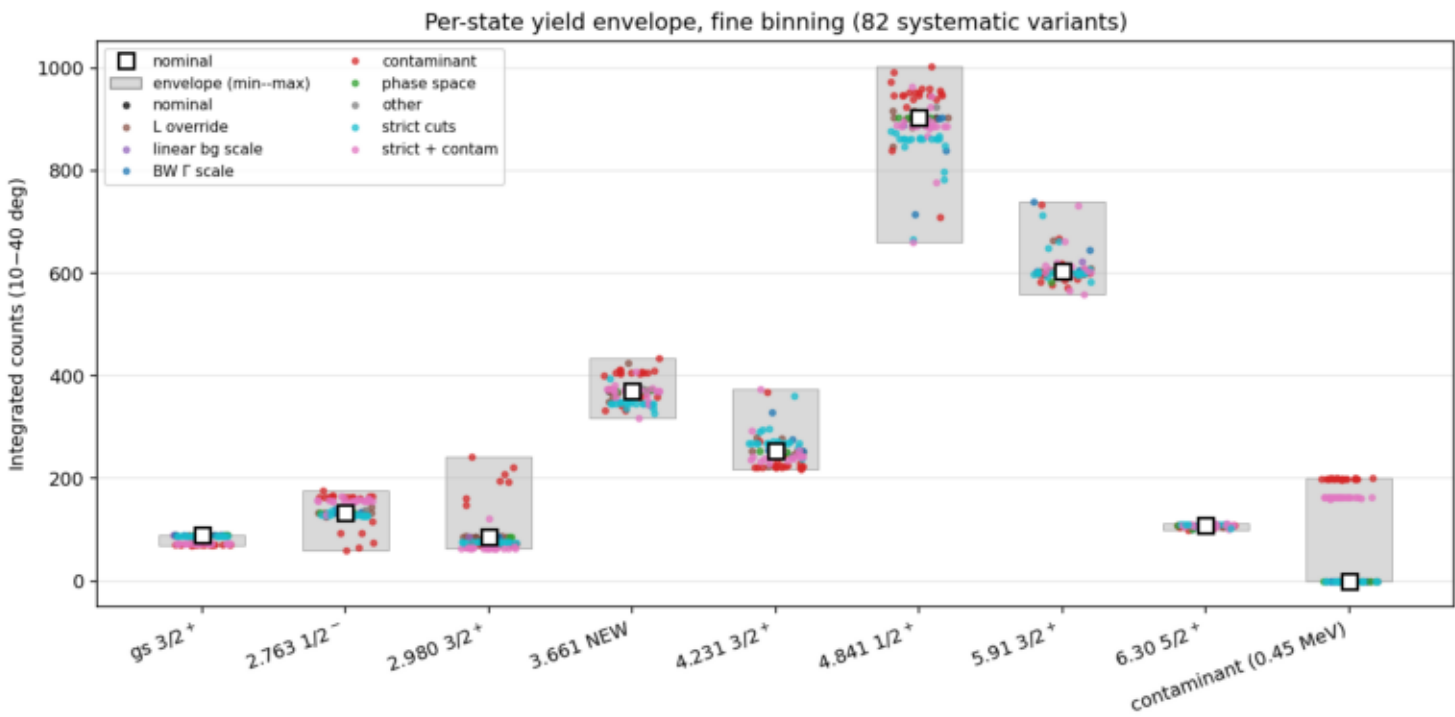


best L (corrected errors):

| Ex    | best L      | next L     |
|-------|-------------|------------|
| ----- |             |            |
| 2.763 | L=3 (1.41)  | L=2 (1.42) |
| 2.980 | L=3 (2.73)  | L=2 (3.05) |
| 3.661 | *L=2 (1.38) | L=3 (1.45) |
| 4.231 | *L=2 (0.58) | L=3 (1.38) |
| 4.841 | *L=0 (3.09) | L=1 (7.35) |
| 5.91  | L=2 (3.55)  | L=0 (3.73) |
| 6.30  | L=0 (0.34)  | L=3 (0.40) |

- $\chi^2/\nu$  refreshed after the error fix.
- All best-L picks & tie flags UNCHANGED;
  - $\chi^2/\nu$  rose  $\times 1.3$ – $1.5$  (every L scales together).

- 4.231  $3/2^+$  / 4.841  $1/2^+$ : decisive (L=2 d-wave; L=0 s-wave) — next L worse by factor  $\gtrsim 2$ ; match the PDF.
- 3.661 (new): L=2 over L=3 (1.38 vs 1.45) — not decisive; possible sd-pf intruder.
- 2.763  $1/2^-$ : three-way near-tie (L=3/2/1); AD too flat to separate.
- 2.980  $3/2^+$ : no template fits well; 17.5° AD peak not reproduced; fit wants  $E_0 \sim 120$  keV above the SM prior.
- 6.30  $5/2^+$ : L=0 has best  $\chi^2/\nu$  but is FORBIDDEN ( $0^+ \otimes s_{1/2} = 1/2^+$ ); allowed L=2/3 are tied.
- 5.91  $3/2^+$ : L=2 marginal over L=0; AD poorly constrained past 30°.



## Integrated-yield envelope:

| state     | -%  | +%   |
|-----------|-----|------|
| -----     |     |      |
| g.s. 3/2+ | -23 | +0   |
| 2.763     | -56 | +32  |
| 2.980     | -28 | +179 |
| 3.661     | -15 | +17  |
| 4.231     | -14 | +48  |
| 4.841     | -27 | +11  |
| 5.91      | -8  | +22  |
| 6.30      | -9  | +3   |

- Largest model-dependence: 2.763 & 2.980 (contaminant degeneracy).
- 6.30 robust (<10%).

# Summary

- Per-state absolute  $d\sigma/d\Omega$  for 3 bound + 7 unbound  $^{17}\text{C}$  states from  $^{16}\text{C}(d,p)$  at 11.77 MeV/u.
- Absolute scale validated vs  $^{16}\text{C}(d,d)$  elastic; normalization uncertainty =  $\sim 12\%$  OMP spread (global).
- Error model corrected: per-point stats no longer double-counted; statistical + systematic + scale layers separated.
- L assignments robust to the fix: 4.231 (L=2) & 4.841 (L=0) decisive; 3.661/2.763/5.91 ambiguous; 2.980 & 6.30 problematic.
- Deliverable (this repo): yields,  $d\sigma/d\Omega$ , systematic envelope. Final L /  $C^2S$  done downstream (CDCC-grade).

# Open items

- Fold the  $\sim 12\%$  elastic-normalization scale into the published  $d\sigma/d\Omega$  / yield bands (common factor).
- Continuum-form tests (quadratic/exp bg, PS-shape) to show the 0.45 MeV contaminant is not bg flexibility.
- Physically identify the 0.45 MeV excess (vertex\_z,  $\theta_{lab}$  fingerprint of a non-D<sub>2</sub> channel?).
- CDCC / pole-residue treatment for publication-grade absolute  $C^2S$ .

Refs: Pereira-Lopez et al., PLB 811 (2020) 135939; Movilla & Ayyad (2026); Pang et al., arXiv:1606.01507 (DA1p).



# Backup: full $\chi^2/\nu$ grid

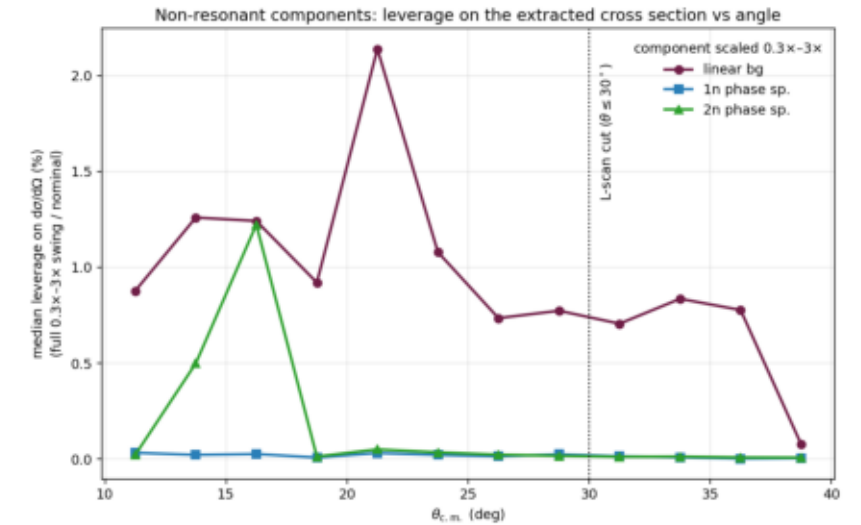
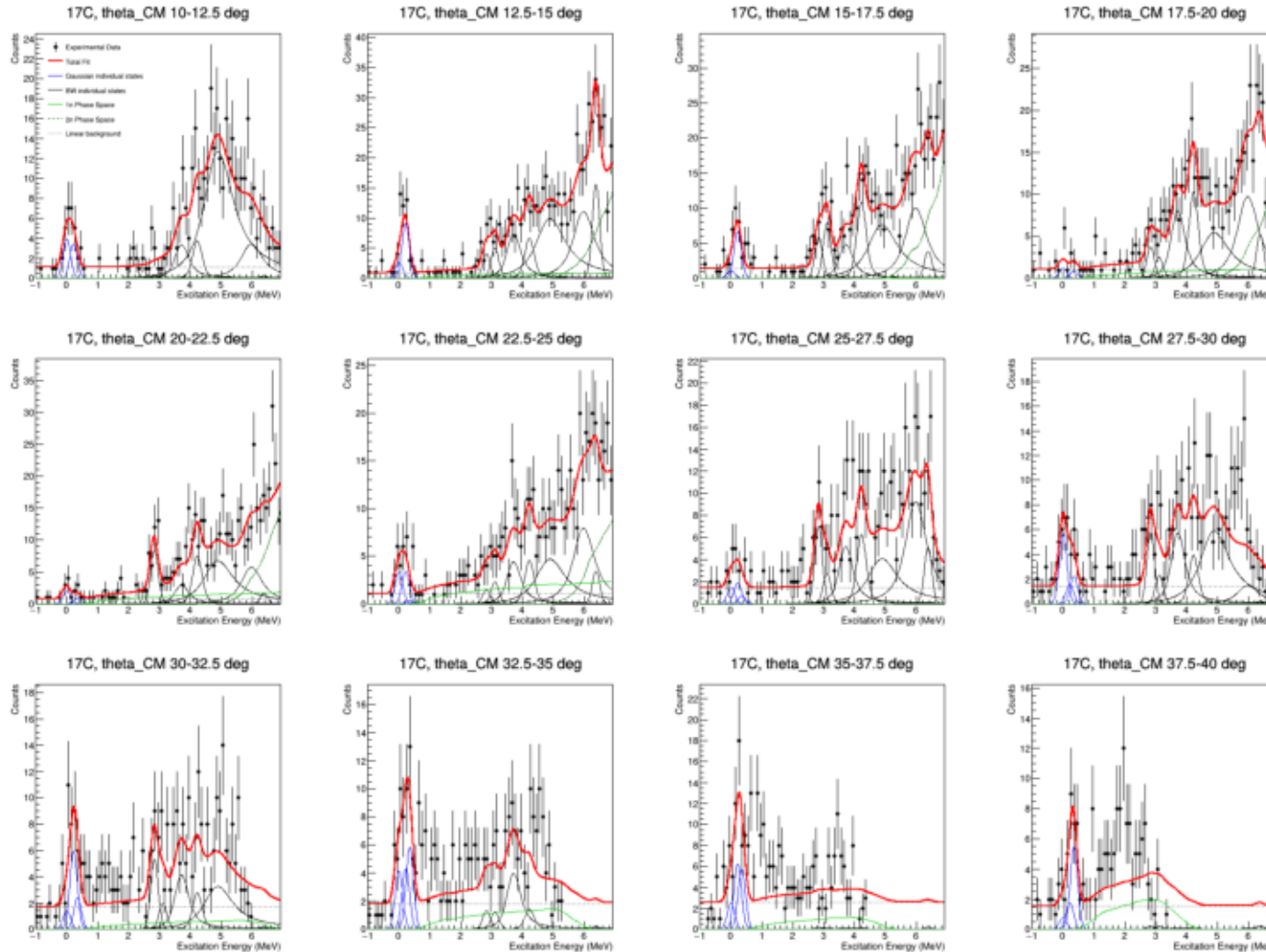
*L values*

| state        | L=0  | L=1    | L=2    | L=3   | best          |
|--------------|------|--------|--------|-------|---------------|
| -----        |      |        |        |       |               |
| 2.763 1/2-   | 3.27 | 1.61   | 1.42 * | 1.41  | L=3 (tie L=2) |
| 2.980 3/2+   | 3.83 | 3.66   | 3.05 * | 2.73  | L=3           |
| 3.661 new    | 3.68 | 4.35 * | 1.38   | 1.45  | L=2 (tie L=3) |
| 4.231 3/2+   | 2.87 | 2.12 * | 0.58   | 1.38  | L=2           |
| 4.841 1/2+ * | 3.09 | 7.35   | 9.23   | 11.53 | L=0           |
| 5.91 3/2+    | 3.73 | 4.25 * | 3.55   | 5.10  | L=2 (tie L=0) |
| 6.30 5/2+ *  | 0.34 | 0.72   | 0.41   | 0.40  | L=0 (tie L=3) |

\* = best L. Fits on theta\_cm <= 30 deg, corrected stat errors (2026-05-30).  
A tie = runner-up within 0.3 in chi2/nu (shapes indistinguishable at current statistics).

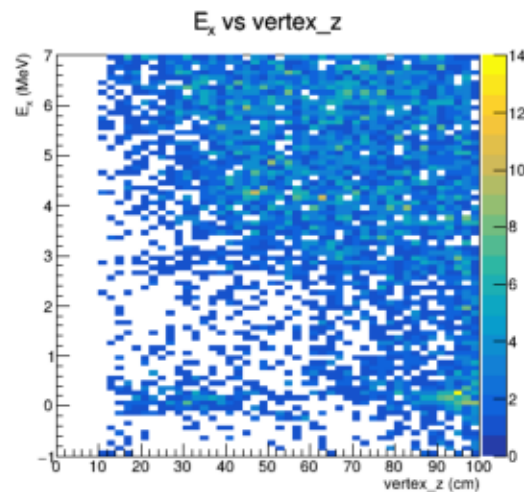
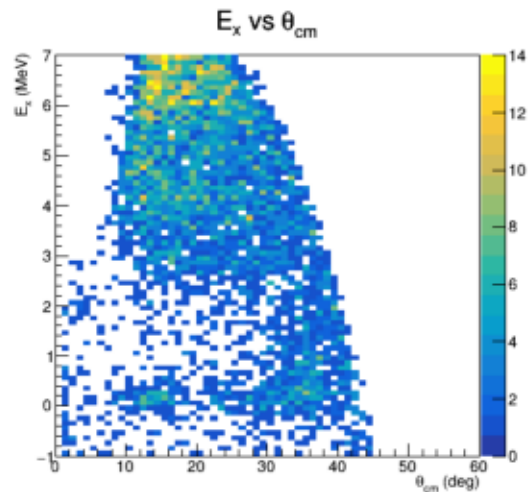
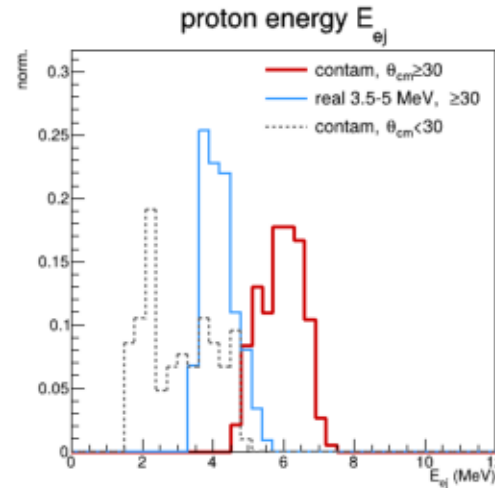
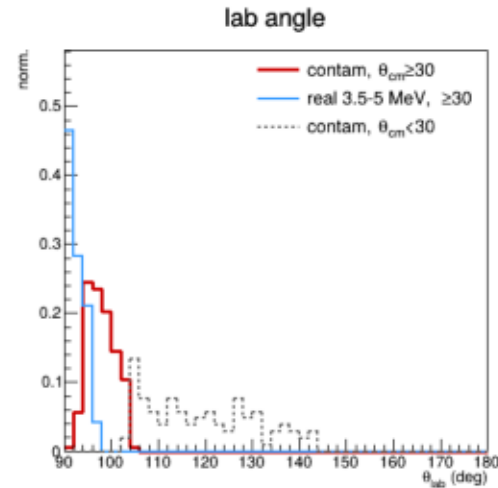
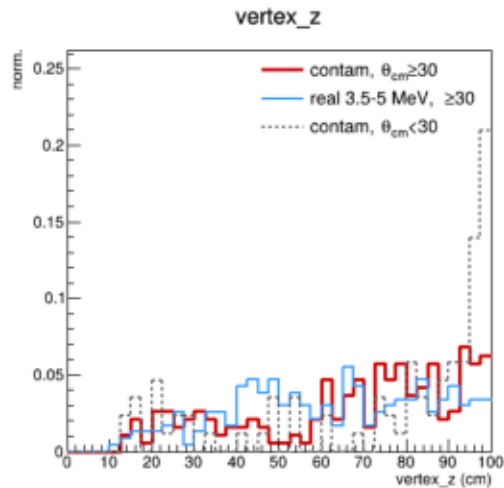
# Backup: non-resonant background treatment

background



- In the fit: 1n PS + linear bg + 2n PS, amplitudes free in EVERY angular bin (PS templates rebuilt per bin).
- Left: per-bin  $E_x$  spectra — peak-dominated forward; from  $\sim 30^\circ$  the transfer peaks fade and the continuum dominates.
- Right: scaling each term  $0.3\times\text{--}3\times$  moves  $d\sigma/d\Omega$  by  $<2\%$  (bg),  $\sim 0\%$  (PS) — flat in angle, no rise past  $30^\circ$ .
- Also in the 82-variant envelope; L-scan uses  $\theta \leq 30^\circ$ . Back-angle limit is stats/efficiency, not bg.

# Backup: the 0.45 MeV contaminant is a low-energy proton background



- Back-angle feature:  $\sim 87\%$  of its fitted yield at  $\theta_{cm} \geq 30^\circ$  (contam\_vs\_angle).
- Fingerprint (47 runs, 56k events): contaminant protons have  $E_{ej} \approx 2.5$  MeV vs  $\sim 5.4$  MeV for genuine transfer at the same back angles.
- Same  $\sim 2.5$  MeV forward too — a fixed low-energy population.
- A real 0.45 MeV  $^{17}\text{C}$  state would give MORE energetic protons than the  $E_x \approx 4$  reference, not half  $\Rightarrow$  these don't obey  $^{16}\text{C}(d,p)$  kinematics.
- Vertex  $\sim 9$  cm downstream (62.7 vs 53.4 cm).
- $\Rightarrow$  low-energy proton background, NOT a  $^{17}\text{C}$  level; a back-angle  $E_{ej}$ /vertex cut would remove it.

## 0.45 MeV contaminant fingerprint

contam-back events: 192

reference-back events: 237

Red vs blue: does the 0.45 MeV back-angle

sample look like real transfer in

vertex\_z /  $\theta_{lab}$  /  $E_{ej}$ ?

Pad 4 ( $E_x$  vs  $\theta_{cm}$ ): a slanted ridge

crossing  $\sim 0.45$  MeV at back angles = ghost.